

Quiz (Habanero)

- Q1.** A forest's average temperature t and rainfall r must satisfy $t + 2r > 15$ and $t - r < 3$. Which point is a feasible solution?
- (A) (10, 2)
(B) (12, 1)
(C) (8, 4)
(D) (6, 6)
(E) (4, 8)
- Q2.** A region's ecosystem can support a maximum of x trees and y animals if $2x + y < 50$ and $x + 2y < 40$. What is the maximum number of trees?
- (A) 15
(B) 20
(C) 25
(D) 30
(E) 35
- Q3.** The temperature T and humidity H in a greenhouse must satisfy $T + H > 20$ and $T - 2H < 10$. Which of the following is a feasible solution?
- (A) (15, 5)
(B) (12, 8)
(C) (10, 10)
(D) (8, 12)
(E) (6, 14)
- Q4.** A city's air quality index A and pollution level P must satisfy $A + 3P < 100$ and $2A - P > 20$. What is the minimum air quality index?
- (A) 20
(B) 25
(C) 30
(D) 35
(E) 40
- Q5.** The amount of x liters of water and y liters of fertilizer used in a farm must satisfy $x + 2y > 15$ and $3x - y < 30$. Which of the following is a feasible solution?
- (A) (10, 2)
(B) (12, 1)
(C) (8, 4)
(D) (6, 6)
(E) (4, 8)
- Q6.** A company's carbon footprint C and energy consumption E must satisfy $C + 2E < 150$ and $E - 2C > 10$. What is the maximum energy consumption?
- (A) 50
(B) 60
(C) 70
(D) 80
(E) 90
- Q7.** The number of x trees planted and y animals protected in a conservation area must satisfy $x + y > 20$ and $2x - y < 15$. Which of the following is a feasible solution?
- (A) (10, 10)
(B) (12, 8)
(C) (8, 12)
(D) (6, 14)
(E) (4, 16)
- Q8.** The temperature T and precipitation P in a region must satisfy $T + P > 25$ and $T - 2P < 15$. What is the minimum temperature?
- (A) 10
(B) 12
(C) 15
(D) 18
(E) 20

Open Ended Questions (FRQ)

- Q9.** A city's water supply system has two sources: a river and a lake. The amount of x gallons of water from the river and y gallons of water from the lake must satisfy $x + 2y > 1000$ and $3x - y < 2000$. Find the maximum amount of water that can be supplied.
- Q10.** A farmer needs to plant x acres of corn and y acres of wheat. The total area available is 100 acres, and the farmer has 1000 to spend on seeds. The cost of seeds for corn is 10 per acre, and the cost of seeds for wheat is 20 per acre. If the profit from corn is 50 per acre and the profit from wheat is 100 per acre, find the number of acres of each crop that will maximize the profit, given that $x + y < 100$ and $10x + 20y < 1000$.

Chef's Tips

- Tip 1: Encourage students to visualize the constraints as a system of inequalities.
- Tip 2: Emphasize the importance of understanding the context of the problem.
- Tip 3: Remind students to check their units and ensure they are consistent throughout the problem.